

# SI Unit

| Quantity            | SI Unit  |
|---------------------|----------|
| Length              | meter    |
| Mass                | kilogram |
| Time                | second   |
| Temperature         | Kelvin   |
| Electric Current    | Ampere   |
| Luminous Intensity  | Candela  |
| Amount of Substance | mole     |

SI symbol

m

kg

s

°K

A

mol

E = exponent of ten  
prefixes:

| Prefixes for Powers of Ten |        |              |           |        |              |
|----------------------------|--------|--------------|-----------|--------|--------------|
| Power                      | Prefix | Abbreviation | Power     | Prefix | Abbreviation |
| $10^{-24}$                 | yocto  | y            | $10^3$    | kilo   | k            |
| $10^{-21}$                 | zepto  | z            | $10^6$    | mega   | M            |
| $10^{-18}$                 | atto   | a            | $10^9$    | giga   | G            |
| $10^{-15}$                 | femto  | f            | $10^{12}$ | tera   | T            |
| $10^{-12}$                 | pico   | p            | $10^{15}$ | peta   | P            |
| $10^{-9}$                  | nano   | n            | $10^{18}$ | exa    | E            |
| $10^{-6}$                  | micro  | $\mu$        | $10^{21}$ | zetta  | Z            |
| $10^{-3}$                  | milli  | m            | $10^{24}$ | yotta  | Y            |
| $10^{-2}$                  | centi  | c            |           |        |              |
| $10^{-1}$                  | deci   | d            |           |        |              |

chain link conversion:  $1 \text{ min} \cdot 1 = \frac{60 \text{ s}}{1 \text{ min}} = 60 \text{ s}$

conversion factor: a ratio of unit that is equal to unity  $\frac{60 \text{ s}}{1 \text{ min}}$

conversion of systems

$$1 \text{ mile} = 1609 \text{ m} = 1.609 \text{ km}$$

$$1 \text{ ft} = 0.3048 \text{ m} = 30.48 \text{ cm}$$

$$1 \text{ m} = 39.37 \text{ in.} = 3.281 \text{ ft}$$

$$1 \text{ in.} = 0.0254 \text{ m} = 2.54 \text{ cm}$$

$$1 \text{ lb} = 0.465 \text{ kg}$$

$$1 \text{ oz} = 28.35 \text{ g}$$

$$1 \text{ slug} = 14.59 \text{ kg}$$

$$1 \text{ day} = 24 \text{ hours} = 24 * 60 \text{ minutes} = 24 * 60 * 60 \text{ seconds}$$

## meter definition



The meter is the length of the path traveled by light in a vacuum during a time interval of  $1/299\,792\,458$  of a second.

**significant figure:** A digit in a number that carry meaningful information about the precision and resolution of measurement.